

CPU Inference Optimization & Deployment

Story Dataset AI System - Final Engineering Report

Final status	COMPLETE
Task 1 deployment	Qwen3-4B Q4_K_M + hierarchical retrieval
Task 2 CPU deployment	TF-IDF + LinearSVC
CPU target	AMD Ryzen 5 4600H, ~16 GB RAM
Runtime	llama.cpp b11146, CPU-only

Optimization principle: Measure -> Profile -> Optimize -> Re-measure -> Quality regression check -> Keep or reject.

This report records both successful and rejected optimizations. The final architecture is based on measured quality, latency, memory, and reliability rather than architectural uniformity.

1. Executive Summary

Task 3 focused on making the conversational QA system and genre classifier practical on CPU. The work began with end-to-end baselines, profiled the real bottlenecks, then tested planner optimization, streaming, context reduction, reranking, thread tuning, quantized inference, LoRA conversion, and lightweight classification.

Area	Best / Final Result
Task 1 original mean latency	37.472 s
Task 1 after hybrid planner	15.688 s (58.1% lower; 2.39x speedup)
Task 1 streaming Top-5	6.743 s mean E2E TTFT; 11.511 s mean full response
Task 1 peak llama-server Working Set	~8.32 GiB
Best Task 2 model quality	Qwen3-4B QLoRA: 65.00% Accuracy, 61.97% Macro F1
Qwen GGUF + LoRA CPU	16.00% Accuracy, 13.34% Macro F1, 25.428 s/story
Final Task 2 CPU choice	TF-IDF + LinearSVC: 57.50% Accuracy, 52.02% Macro F1, ~0.151 ms/story

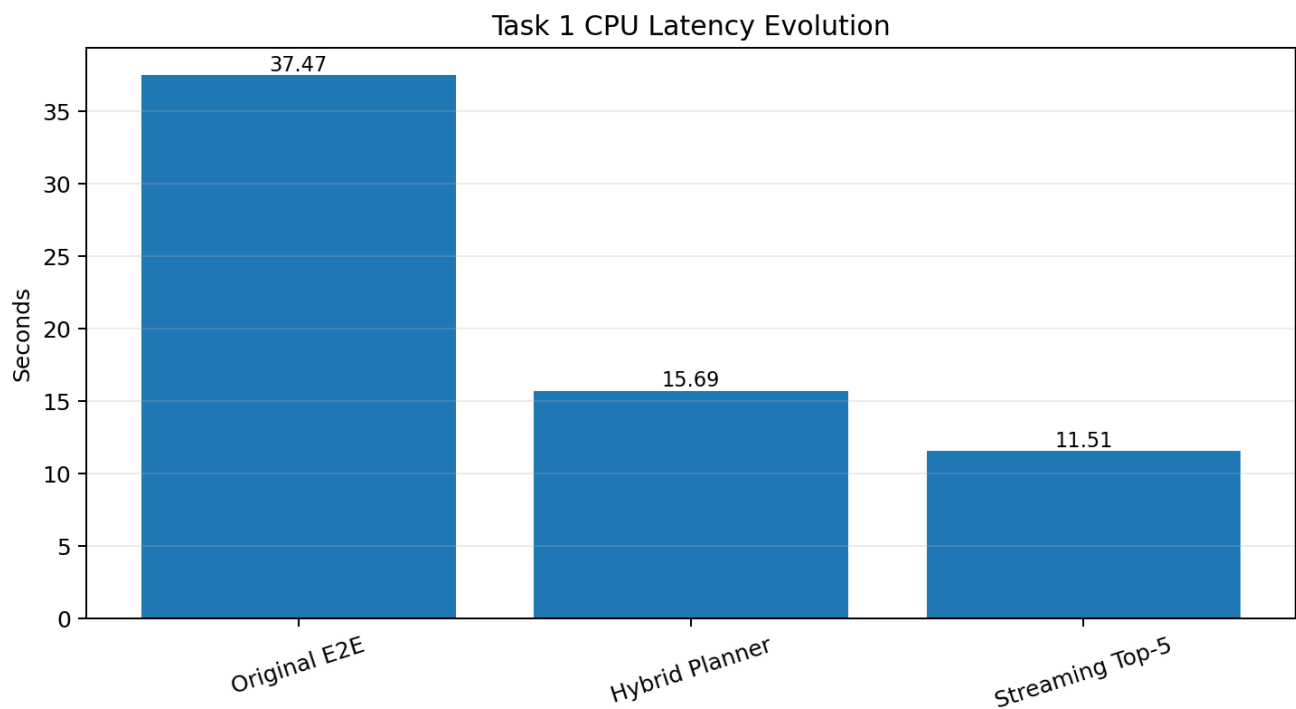


Figure 1. Task 1 latency evolution. The controlled hybrid-planner benchmark produced the largest accepted end-to-end improvement; streaming Top-5 further improved the steady-state user experience.

2. CPU Environment and Runtime Verification

The deployment benchmarks were executed using the Windows x64 CPU build of llama.cpp. Runtime device discovery with **llama-server --list-devices** returned **Available devices: (none)**, so no accelerator device was available for model, KV-cache, or tensor offloading.

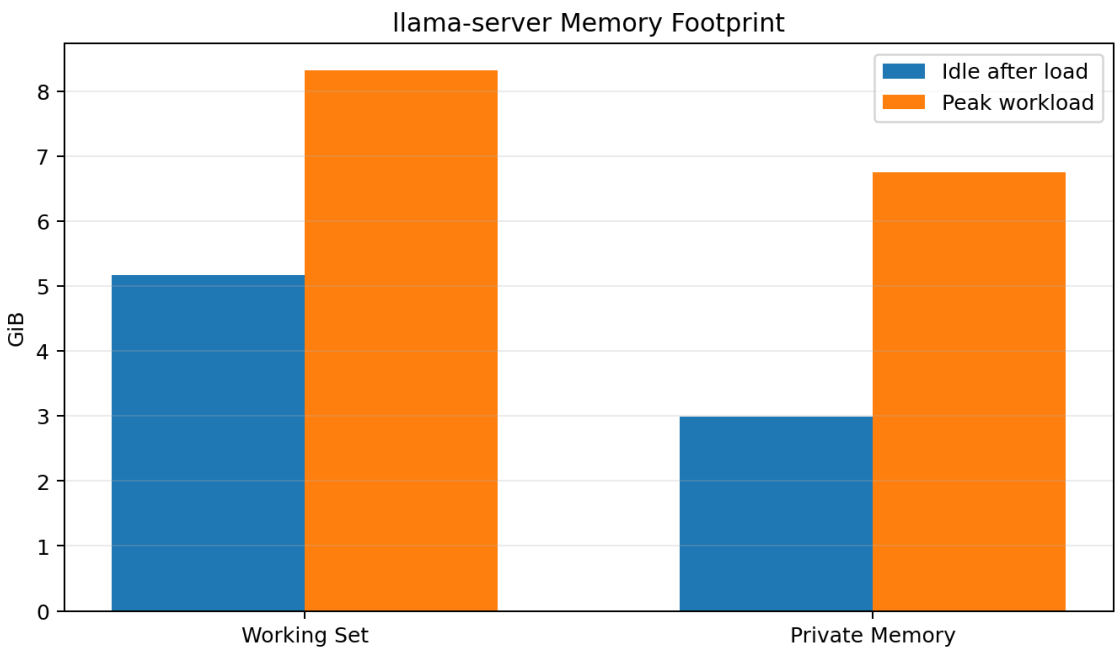
Component	Configuration
CPU	AMD Ryzen 5 4600H with Radeon Graphics
Physical / logical cores	6 / 12
System RAM	15.37 GiB (~16 GB)
Runtime	llama.cpp b11146, Windows x64 CPU build
Accelerators	None detected
Task 1 LLM	Qwen3-4B Q4_K_M GGUF
Context window	8192 tokens

2.1 Thread Tuning

Configuration	Mean	Median	P95	Generation rate
Automatic (-1)	3.393 s	2.944 s	5.274 s	10.82 tok/s
6 threads	3.636 s	3.247 s	5.656 s	10.17 tok/s
12 threads	4.130 s	3.599 s	6.428 s	8.92 tok/s

Explicit thread counts did not improve performance, so llama.cpp automatic thread configuration was retained.

2.2 Runtime Memory



Metric	Idle after model load	Peak during Task 1
Working Set	5.17 GiB	8.32 GiB
Private Memory	2.99 GiB	6.76 GiB

Peak memory remained within the available ~15.37 GiB system RAM.

3. Task 1 - Profiling and Planner Optimization

The original 20-query QA benchmark contained 15 content-QA and 5 long-story-QA queries. Model and embedding startup were excluded from per-request timing.

Metric	Original CPU baseline	After hybrid planner
Mean	37.472 s	15.688 s
Median	41.324 s	14.603 s
P95	46.993 s	27.931 s
Minimum	10.120 s	2.680 s
Maximum	50.866 s	34.896 s
Content QA mean	35.044 s	16.278 s
Long-story QA mean	44.754 s	13.916 s

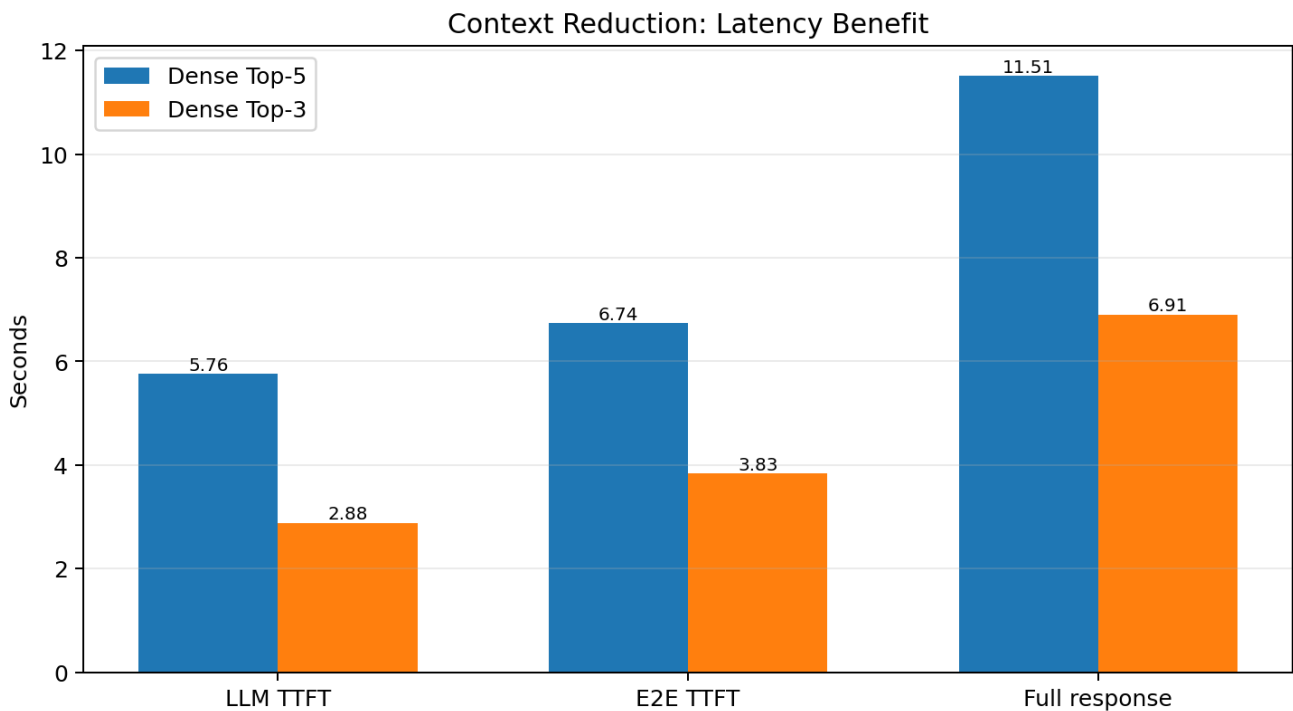
Profiling showed planner latency around 8-9 seconds, warmed story/evidence retrieval around 0.01-0.03 seconds, and generation around 6-15 seconds. Therefore, the planner and Qwen inference - not FAISS - were the main optimization targets.

The final hybrid planner uses deterministic routing only when safe and falls back to Qwen otherwise. On the 50-query planner benchmark it achieved **100% intent accuracy**, **100% retrieval-strategy accuracy**, 4.270 s average planner latency, 6.858 s P50, and 9.024 s P95.

Accepted result: 58.1% lower mean end-to-end latency and approximately 2.39x mean speedup while preserving planner benchmark behavior.

4. Streaming and Context-Size Optimization

Streaming separates pre-generation work from model TTFT and full-response latency. For the production Dense Top-5 evidence configuration, mean pre-generation latency was 0.980 s, mean LLM TTFT 5.763 s, mean end-to-end TTFT 6.743 s, and mean full-response latency 11.511 s.



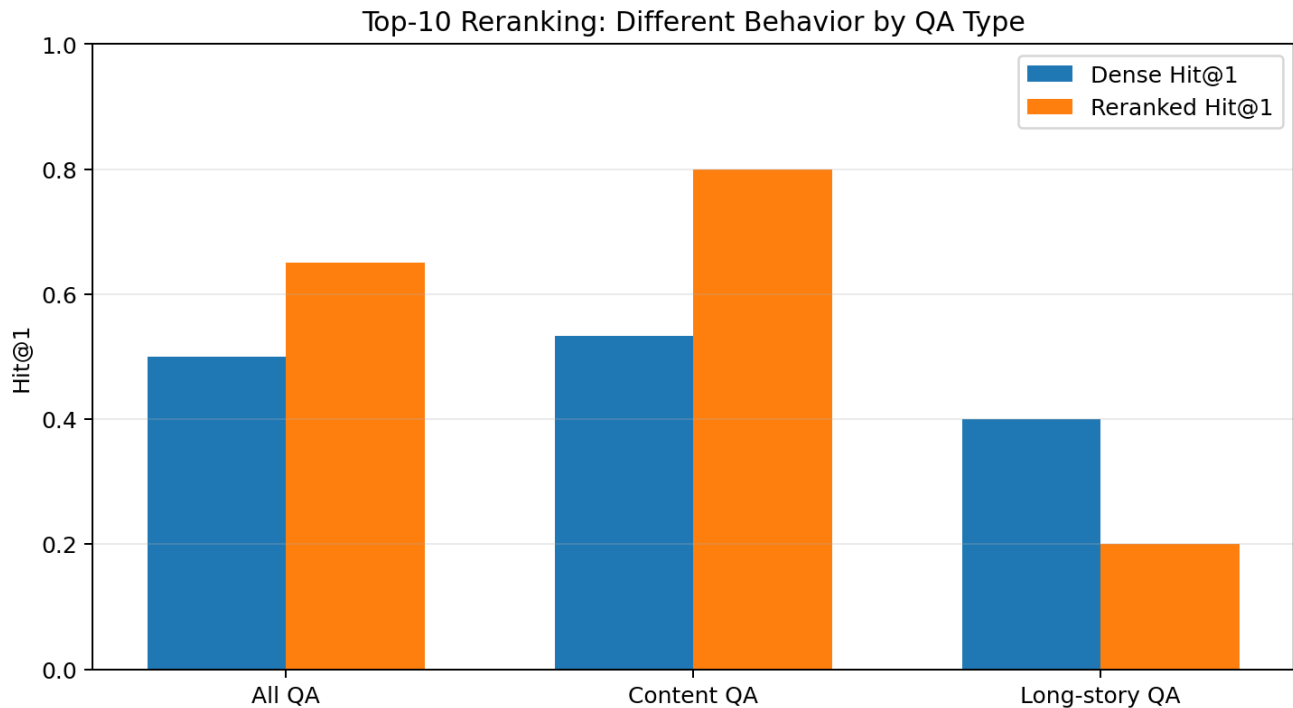
Metric	Top-5	Top-3	Change
Pre-generation mean	0.980 s	0.946 s	-3.5%
LLM TTFT mean	5.763 s	2.884 s	-50.0%
LLM TTFT P50	5.355 s	1.982 s	-63.0%
LLM TTFT P95	12.823 s	5.458 s	-57.4%
E2E TTFT mean	6.743 s	3.831 s	-43.2%
Full-response mean	11.511 s	6.908 s	-40.0%
Full-response P50	11.414 s	3.825 s	-66.5%
Full-response P95	19.499 s	19.058 s	-2.3%

Despite the large latency gain, fixed Top-3 was rejected. Q027 and Q036 lost required evidence; Q039 became incomplete; Q043 produced an incorrect relationship; and Q046 incorrectly described Project Echo's objective.

Decision: Dense Top-5 remains the production evidence configuration. Top-3 is retained only as a measured latency/quality ablation.

5. Evidence Reranking Experiments

A lightweight **cross-encoder/ms-marco-MiniLM-L6-v2** reranker was evaluated to test whether a wider candidate pool could be compressed to a smaller, higher-quality context.



5.1 Top-5 Candidate Pool

Subset	Method	Hit@1	Hit@3	Hit@5	MRR
All QA	Dense	0.500	0.800	0.950	0.660
All QA	Reranked	0.650	0.850	0.950	0.762
Content QA	Dense	0.533	0.933	1.000	0.717
Content QA	Reranked	0.800	1.000	1.000	0.889
Long QA	Dense	0.400	0.400	0.800	0.490
Long QA	Reranked	0.200	0.400	0.800	0.380

5.2 Expanded Top-10 Candidate Pool

The expanded dense pool contained 145 candidate rows and 46 judged supporting chunks, including 9 newly discovered supporting chunks. Dense retrieval achieved **Hit@10 = 100%** on the 20-query QA benchmark.

Subset	Method	Hit@1	Hit@3	Hit@5	Hit@10	MRR
All QA	Dense	0.500	0.800	0.950	1.000	0.666
All QA	Reranked	0.650	0.800	0.850	1.000	0.758
Content	Dense	0.533	0.933	1.000	1.000	0.717
Content	Reranked	0.800	0.933	1.000	1.000	0.883
Long	Dense	0.400	0.400	0.800	1.000	0.515
Long	Reranked	0.200	0.400	0.400	1.000	0.381

CPU reranking cost was small: 5.527 s total, 0.276 s/query, and 0.0381 s/pair. Quality, not compute, was the rejection reason.

6. Why the Reranker Was Rejected

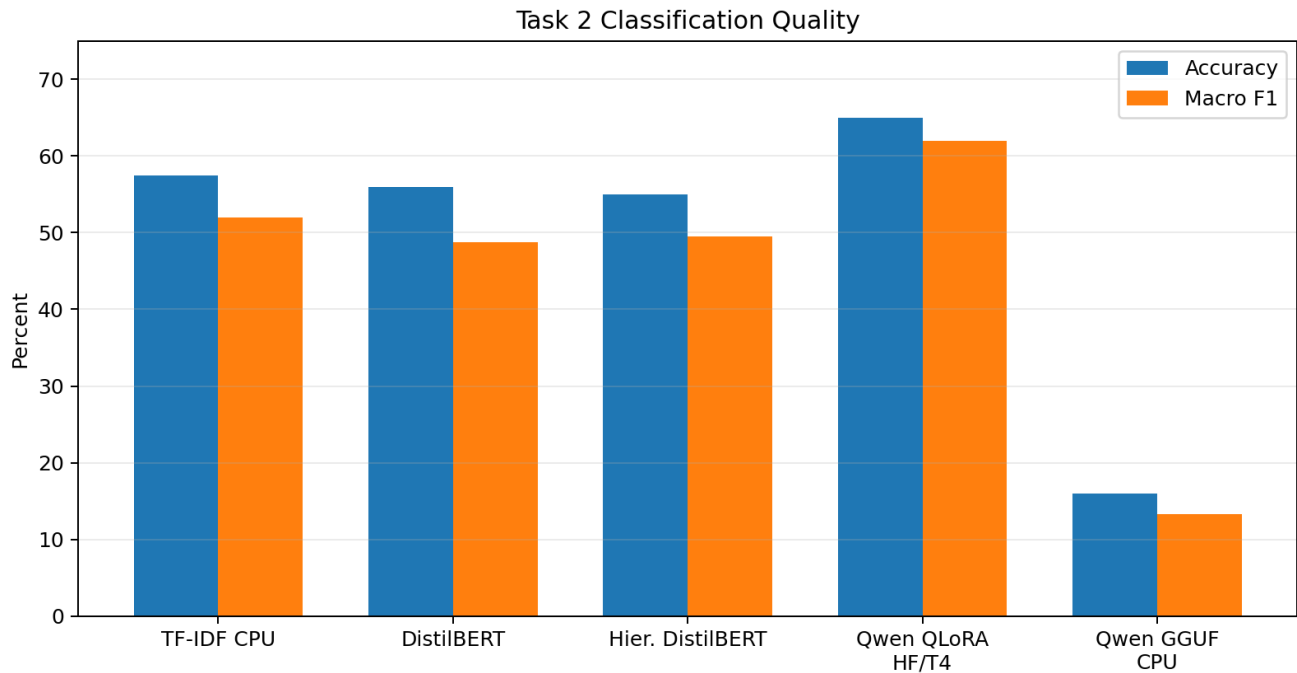
The failure analysis showed that generic semantic relevance is not identical to answer-bearing relevance in long narrative stories.

Query	Required evidence	Dense rank	Reranked rank	Observation
Q043	Archer's relationship to Red Rose	8	6	Topical passages outranked the exact relationship evidence.
Q044	Amir is an accountant at a prestigious law firm	1	8	Exact opening evidence was strongly demoted.
Q046	Project Echo creates AI surpassing human intelligenc	5	9	Related AI passages outranked the answer-bearing passage

Engineering conclusion: High semantic relevance != answer-bearing evidence. Therefore, the universal strategy Retrieve Top-10 -> CrossEncoder -> always send Top-3 was rejected.

The reranker remains a useful ablation: it improves ordinary content-QA ordering, but a safer answer-evidence selector would be required before using it as a universal context-compression mechanism.

7. Task 2 - Classification Reference Results



Model	Accuracy	Macro F1	Deployment note
TF-IDF + LinearSVC	57.50%	52.02%	Same 799/200 split; ~0.151 ms/story
DistilBERT 512	56.00%	48.78%	First 512 tokens
Hierarchical DistilBERT	55.00%	49.49%	Chunk aggregation
Qwen3-4B QLoRA E1	65.00%	61.97%	Best quality; HF/NF4 + PEFT on T4

The reproducible Qwen checkpoint is checkpoint-100 (epoch 1). It achieved 65.00% accuracy, 65.98% Macro Precision, 65.40% Macro Recall, 61.97% Macro F1, 61.75% Weighted F1, a 2.50% invalid-output rate, and 2.727 s/story average T4 latency.

A later training attempt reached step 196 but was interrupted before a checkpoint was saved; it is not used in reported results.

7.1 Task 1 Regression Check After Genre LoRA

Using the same quantized base, full-story context, prompts, tokenizer, and deterministic generation, no factual QA regression was observed on Q027-Q046 (20 questions). Manual semantic review found the expected answer preserved on all 20 tested questions. This is intentionally scoped to that QA subset.

8. QLoRA to GGUF CPU Deployment

The PEFT checkpoint was converted with llama.cpp's LoRA converter to an F16 GGUF adapter. The conversion exported 504 tensors. The original adapter safetensors was ~132.2 MB; the F16 GGUF adapter was ~66.1 MB. F16 was chosen to avoid adding LoRA quantization as another quality variable.

The adapter loaded successfully with the Q4_K_M base in llama.cpp. CPU_REPACK fallback warnings appeared for LoRA tensors, but the server loaded and served requests.

8.1 Initial CPU Run - Preprocessing Failure

Metric	Initial run
Accuracy	12.50%
Macro F1	8.61%
Weighted F1	11.02%
Invalid outputs	29/200 (14.50%)
Truncated inputs	29/200 (14.50%)
Mean latency	25.503 s/story
P95 latency	40.796 s/story

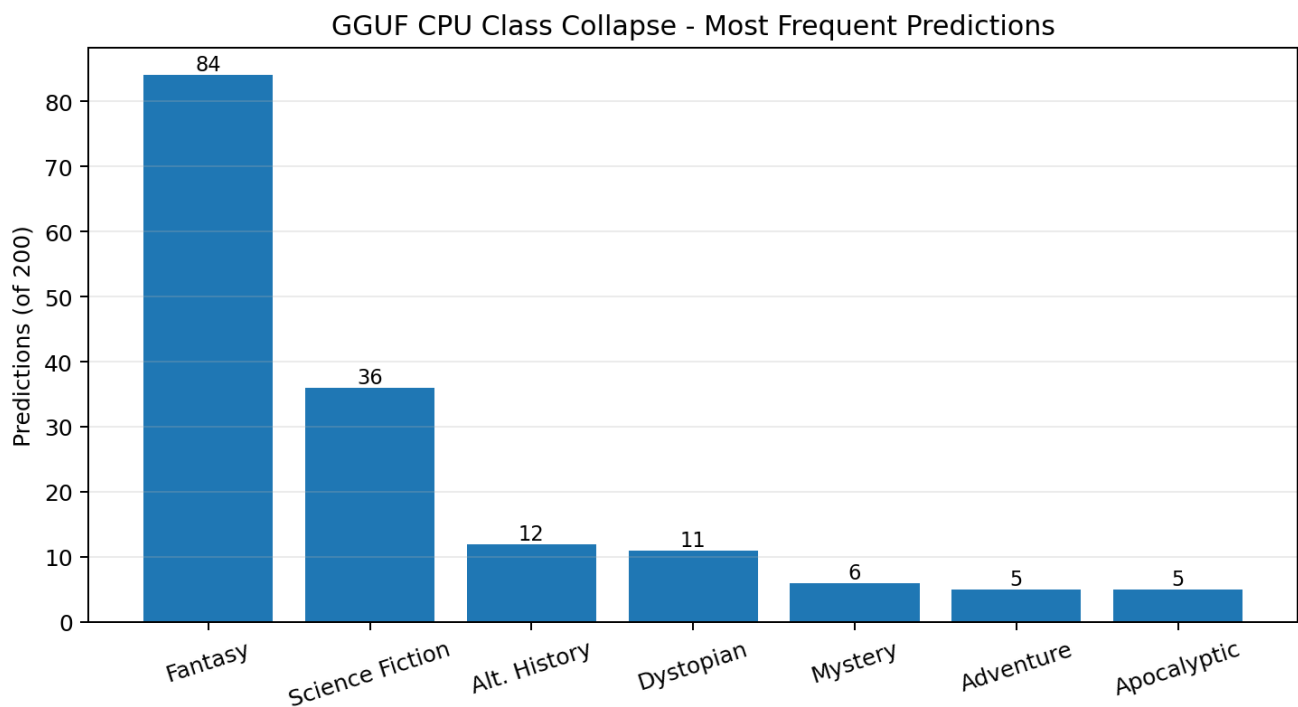
Inspection showed an exact relationship: all 171 non-truncated prompts produced valid labels, while all 29 prompts truncated at the complete-sequence level produced invalid outputs. The old code could remove the final Genre instruction and assistant boundary, causing Qwen to continue the story instead of classifying it.

Fix: reserve the system instruction, allowed-label list, final Genre instruction, and assistant boundary; truncate only the story; then rebuild the complete prompt. Only the 29 affected stories were rerun.

9. Corrected Qwen GGUF + LoRA CPU Benchmark

Metric	Corrected result
Validation stories / genres	200 / 99
Accuracy	16.00%
Macro Precision	15.43%
Macro Recall	16.16%
Macro F1	13.34%
Weighted F1	13.21%
Invalid outputs	0/200 (0.00%)
Story-truncated inputs	29/200 (14.50%)
Mean latency	25.428 s/story
Median latency	25.625 s/story
P95 latency	41.397 s/story

The preprocessing correction eliminated invalid outputs completely, proving the truncation diagnosis. However, classification quality remained far below the HF/PEFT reference.



Fantasy and Science Fiction alone accounted for 120/200 predictions (60%), despite the approximately balanced 99-class target distribution. This is severe class collapse.

10. Base-Q4 vs GGUF-LoRA Diagnostic

Twenty validation examples were run on the Q4_K_M base without LoRA and compared with the already-computed Q4_K_M + GGUF LoRA predictions.

Diagnostic	Result
Examples compared	20
Identical predictions	20
Different predictions	0
Agreement	100%

Representative example:

Configuration	Prediction
Expected label	Steampunk Fantasy
HF NF4 + PEFT LoRA	Steampunk Fantasy
Q4_K_M base	Fantasy
Q4_K_M + GGUF LoRA	Fantasy

The adapter was loaded by llama.cpp, so this does not prove technical absence of the LoRA. The supported conclusion is that the converted adapter produced no observable prediction change on the 20-example diagnostic and did not preserve the classifier behavior measured in the HF/PEFT setup.

Potential causes include base quantization differences, adapter conversion/application behavior, runtime implementation differences, or their interaction. Further low-level debugging was not justified because this path failed both the quality and latency objectives.

11. Final Task 2 CPU Deployment Decision

Configuration	Accuracy	Macro F1	Latency
Qwen QLoRA - HF/T4 reference	65.00%	61.97%	2.727 s/story
TF-IDF + LinearSVC - CPU	57.50%	52.02%	~0.151 ms/story
Qwen Q4_K_M + GGUF LoRA - CPU	16.00%	13.34%	25.428 s/story

Compared with the best Qwen QLoRA model, TF-IDF sacrifices 7.5 percentage points of accuracy and 9.95 percentage points of Macro F1, but provides extremely low CPU latency. The Qwen GGUF path is both substantially less accurate and dramatically slower than TF-IDF in the measured deployment.

Final Task 2 CPU choice: TF-IDF + LinearSVC.

The Qwen QLoRA checkpoint remains the best Task 2 model by classification quality; TF-IDF + LinearSVC is the final CPU deployment model because it provides the strongest measured quality/latency trade-off.

12. Final Optimization Decision Matrix

Experiment	Measured outcome	Decision
Qwen Q4_K_M / llama.cpp	CPU LLM deployment feasible for QA	KEEP for Task 1
Hybrid planner	58.1% lower controlled mean E2E latency; planner benchmark preserved	KEEP
Streaming	Improves perceived latency; Top-5 E2E TTFT 6.743 s	KEEP
Dense Top-3	~40% lower mean full response but QA regressions	REJECT as default
Top-5 CrossEncoder	Improves content QA; inconsistent long QA	Do not deploy globally
Dense Top-10	Hit@10 = 100% on 20-query evidence benchmark	Useful candidate strategy
Top-10 -> reranker -> Top-3	Long-story answer evidence can be demoted	REJECT globally
Explicit 6/12 threads	Slower than automatic setting	REJECT; retain auto
Qwen GGUF genre LoRA	16% Accuracy; 13.34% Macro F1; 25.428 s/story	REJECT for Task 2
TF-IDF + LinearSVC	57.5% Accuracy; 52.02% Macro F1; ~0.151 ms/story	SELECT for Task 2 CPU

13. Final CPU Architecture

Task 1 - Conversational QA

User Query -> Hybrid deterministic/LLM planner -> Hierarchical story/evidence retrieval -> Dense Top-5 evidence -> Qwen3-4B Q4_K_M / llama.cpp -> Streaming response

Task 2 - Genre Classification

Story -> TF-IDF (1-2 grams, max_features=50,000) -> LinearSVC -> one of 99 genres

The initial shared-Qwen-base idea was attractive architecturally, but the converted CPU LoRA did not preserve classification quality. The final design therefore prioritizes measured production behavior over architectural uniformity.

14. Engineering Lessons

Measure before optimizing. Profiling identified planner and LLM inference as the real Task 1 bottlenecks; warmed vector retrieval was already cheap.

Latency improvements require regression testing. Top-3 was much faster but lost evidence and degraded answers.

Candidate recall and evidence selection are different. Top-10 dense retrieval reached 100% Hit@10, but safely compressing those candidates remained difficult.

Semantic relevance is not necessarily answer-bearing relevance. The generic CrossEncoder could demote exact factual evidence in long stories.

Deployment conversion must be re-evaluated. The 65% HF/PEFT classifier fell to 16% after the tested GGUF deployment path, even after fixing preprocessing.

Preprocessing bugs can look like model failures. Story-only truncation reduced invalid outputs from 14.5% to 0%.

Failed optimizations are useful engineering results. The rejected reranker and GGUF-LoRA paths directly informed the final architecture.

15. Final Scorecard

Area	Final result
CPU-only verification	Confirmed; llama.cpp reported no accelerator devices
Task 1 LLM	Qwen3-4B Q4_K_M
Task 1 production evidence	Dense Top-5
Task 1 original mean E2E	37.472 s
Task 1 hybrid mean E2E	15.688 s
Task 1 speedup	2.39x
Task 1 streaming mean E2E TTFT	6.743 s
Task 1 streaming mean full response	11.511 s
Peak Working Set	~8.32 GiB
Best Task 2 quality	Qwen QLoRA: 65.00% Acc / 61.97% Macro F1
Task 2 GGUF CPU attempt	16.00% Acc / 13.34% Macro F1 / 25.428 s
Final Task 2 CPU model	TF-IDF + LinearSVC
Final Task 2 CPU quality	57.50% Acc / 52.02% Macro F1
Final Task 2 CPU latency	~0.151 ms/story

FINAL STATUS: TASK 3 COMPLETE

The final system uses Qwen3-4B Q4_K_M for conversational QA and TF-IDF + LinearSVC for CPU genre classification. This is the architecture best supported by the measured quality, latency, memory, and reliability results.